

# 1. Phase 1: Multi-Modal Feature Matrix & Microenvironment Deconvolution

## Analytical Methodology & Rationale

- **What is being done:** Loading preprocessed clinical, transcriptomic, and genomic data and constructing dual feature matrices: an ICI-treated cohort ( $N_{\text{ICI}} = 326$ ) and a full melanoma cohort ( $N_{\text{Full}} = 699$ ) with engineered immune signatures, Macrophage STV ratios, and cell deconvolution metrics.
- **Why we are doing it:** Raw RNA-seq gene expression matrices containing ~19,757 genes suffer from severe dimensionality challenges. Transforming high-dimensional transcriptomics into validated signature scores and cell-type fractions provides interpretable, non-redundant biological features for both response prediction and unsupervised stratification.
- **What question it answers:** How are raw multi-modal datasets harmonised and structured into dual feature matrices to power downstream response-supervised modelling and unsupervised patient stratification?

Phase 1 establishes the foundational dataflow architecture by integrating harmonised multi-modal data across 4 melanoma studies ([Liu 2019](#), [Riaz 2017](#), [Hugo 2016](#), and [TCGA-SKCM](#)). To support distinct analytical requirements across downstream phases, Phase 1 outputs two standardised feature matrices:

## Dual Feature Matrix Dataflow Architecture

1. **ICI-Treated Feature Matrix** ( `feature_matrix.csv` ,  $N_{\text{ICI}} = 326$ ):
  - **Composition:** Comprises immunotherapy-treated patients across 4 studies ([Liu 2019](#), [Riaz 2017](#), [Hugo 2016](#), and ICI-treated [TCGA-SKCM](#) subset) with complete RECIST clinical response labels ( `RESPONSE_BINARY` ).
  - **Downstream Routing:** Powers response-supervised analyses: **Phase 2** (Feature Analysis & Youden Cutoffs), **Phase 5** (Subgroup Predictive Modelling), and **Phase 6** (Clinical Utility & Decision Curve Analysis).
2. **Full Melanoma Feature Matrix** ( `feature_matrix_full.csv` ,  $N_{\text{Full}} = 699$ ):
  - **Composition:** Merges ICI trial cohorts with the complete reference cohort ([TCGA-SKCM](#),  $N = 443$ ), expanding the dataset to capture overall population-level biological heterogeneity.
  - **Downstream Routing:** Powers response-agnostic biological discovery and decision support: **Phase 3** (Unsupervised Patient Stratification & Manifold Projections), **Phase 4** (Phenotype Characterisation & Dynamic Trajectories), and **Phase 7** (3-Arm Decision Support & Treatability Index Scoring).

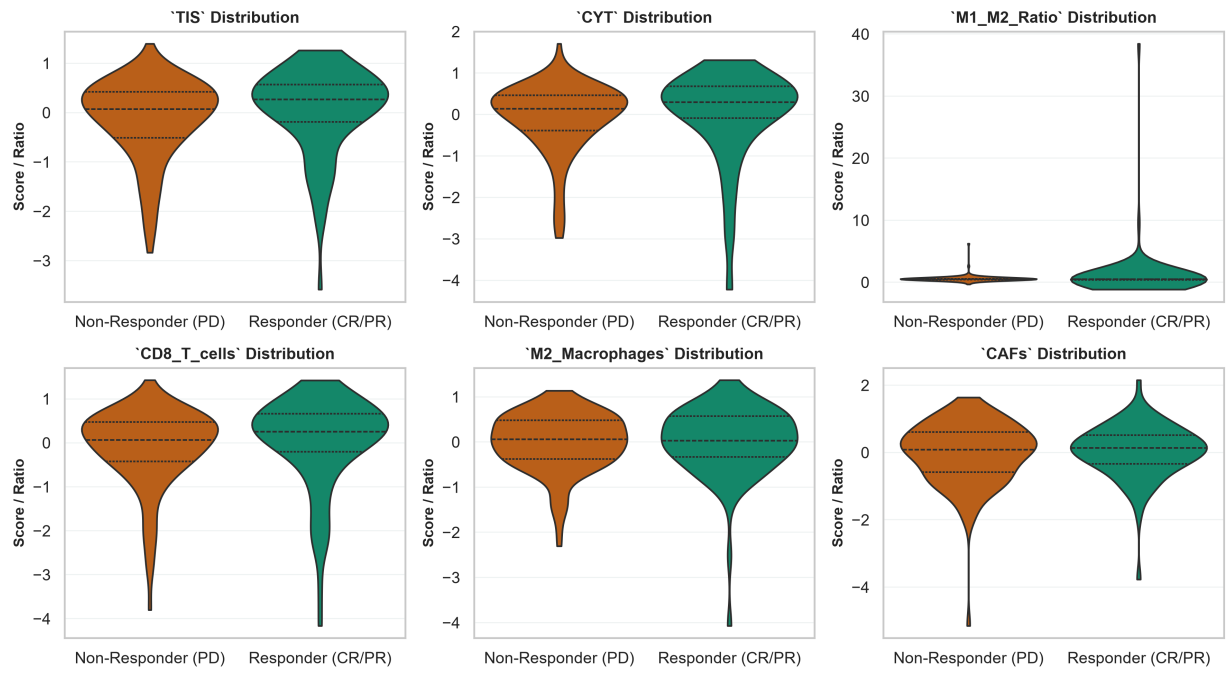
## Biological Feature Engineering & Microenvironment Deconvolution

Rather than evaluating ~19,757 genes independently, Phase 1 projects patient expression profiles onto curated biological axes:

- **Core Immune Signatures:** Tumour Inflammation Signature ( `TIS` ), Cytolytic Index ( `CYT` , mean of `PRF1` and `GZMA` ), Interferon-gamma ( `IFN_gamma` ), and `CD274` ( `PD-L1` ) expression.
- **Macrophage STV** ( `M1_M2_Ratio` ): Computed using a linear Signature Transcript Vector ( $W_g$ , 14,835 genes) to quantify the microenvironmental balance between pro-inflammatory M1 macrophages ( $W_g > 0$ ) and pro-tumour M2 macrophages ( $W_g < 0$ ).
- **Transcriptomic Cell Deconvolution:** Marker-based signature scores estimating the relative infiltration abundance of CD8+ T cells ( `CD8_Tcell` ), CD4+ T cells, NK cells, B cells, M1 Macrophages, M2 Macrophages, and Cancer-Associated Fibroblasts ( `CAFs` ).

## Baseline Biomarker Feature Distributions

### Baseline Immune & Microenvironmental Biomarker Distributions (N=195)



#### Key Takeaways

- **Dual-Matrix Dataflow:** Established a dual dataflow pipeline isolating response-labeled ICI trials ( $N_{ICI} = 326$ ) for predictive modelling while embedding the full cohort ( $N_{Full} = 699$ ) for unsupervised manifold learning.
- **Dimensionality Reduction:** Compressed ~19,757 transcriptomic features into 17 engineered biological signatures (part of a 26-feature multi-modal panel, centered on 9 core baseline biomarkers).
- **M1/M2 Polarisation:** The Macrophage STV score captures stromal microenvironmental suppression that operates independently of total T-cell density.

## Phase 1 Feature Matrix Architecture & Complete Feature Inventory

- **Transcriptomic Features (17):**
  - **Core Immune Signatures (6):**
    1. TIS (Tumour Inflammation Signature)
    2. CYT (Cytolytic Index)
    3. IFN\_gamma (Interferon-gamma signalling)
    4. CD8\_Tcell (*CD8A/B*)
    5. IMPRES
    6. PD\_L1 ( *CD274* ).
  - **Macrophage STV Metrics (4):**
    1. M1\_score
    2. M2\_score
    3. M1\_M2\_Ratio
    4. Macrophage\_STV\_Score (Signature Transcript Vector balance).
  - **Transcriptomic Cell Deconvolution (7):**
    1. CD8\_T\_cells
    2. CD4\_T\_cells
    3. NK\_cells
    4. B\_cells
    5. M1\_Macrophages
    6. M2\_Macrophages
    7. CAFs (Cancer-Associated Fibroblasts).
- **9 Genomic, TMB & Neoantigen Features:**
  - **Driver Mutations (3):**
    1. mut\_BRAF
    2. mut\_NRAS
    3. mut\_NF1 (binary oncogenic driver status).
  - **TMB & Neoantigen Burden (6):**
    1. TMB\_NONSYNONYMOUS
    2. SNV\_NEOANTIGEN
    3. INDEL\_NEOANTIGEN
    4. FUSION\_NEOANTIGEN
    5. SPLICE\_NEOANTIGEN
    6. CTA\_SELF\_NEOANTIGEN .
- **9 Core Baseline Biomarkers (Primary Subset):**
  1. TIS
  2. CYT
  3. PD\_L1
  4. Macrophage\_STV\_Score
  5. CD8\_T\_cells
  6. CD4\_T\_cells
  7. NK\_cells
  8. B\_cells
  9. CAFs (used for primary volcano, Youden ROC, and radar visualisations).